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Surrogate-based Monte Carlo propagation for expensive measurement models: when passive Gaussian processes fail and what a surrogate committee needs

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***Abstract*— Monte Carlo propagation of distributions needs thousands of evaluations of the measurement model, which is infeasible when each evaluation is a fluid dynamics simulation lasting hours. A surrogate trained on about twenty runs makes propagation cheap but adds a rarely evaluated model-form uncertainty component. On a benchmark with a controllable domain, a passively sampled Gaussian process becomes no better than a constant predictor once the run spacing is large. A committee of structurally different surrogates recovers accuracy only when one member represents the dominant response component; paired 30-seed ablations show that the run-selection metric does not matter. Aggregation is decisive: uniform averaging failed in all 30 trials, while cross-validation error weighting succeeded whenever the matching member was retained. The weighted committee reproduced a reference Monte Carlo evaluation to within 0.7 % and supplies a model-form variance share and probability-box bounds for the uncertainty budget.**

***Keywords*—Measurement uncertainty; Monte Carlo method; Surrogate model; Active learning; Model-form uncertainty**

I. INTRODUCTION

Evaluating measurement uncertainty by the propagation of distributions, following Supplement 1 to the Guide to the Expression of Uncertainty in Measurement (GUM S1) [1], [2], requires the measurement model $Y = f(\mathbf{X})$ to be evaluated once per Monte Carlo trial. For many quantities of practical interest, f is itself a numerical simulation. One example is the concentration downstream of an accidental ammonia release in an industrial hall [3]. A single computational fluid dynamics (CFD) run of such a release takes hours to days, while the standard error of a Monte Carlo estimate falls only as $M^{-1/2}$ in the number of trials M. A practical budget is often fewer than twenty runs.

The usual remedy is a *surrogate* (emulator) trained on those runs and used in place of f inside the Monte Carlo loop [4], [5]. This trades an intractable cost for a new contribution to the uncertainty budget: the error of the surrogate itself. From a metrological point of view, this contribution must be evaluated and reported, not assumed to be negligible. The standard surrogate, Gaussian process (GP) regression [6], [7], provides a predictive variance, but that variance is conditional on the GP being an adequate model. Where it is not, the GP can be confidently wrong [8].

Committee-based schemes respond by training several structurally different surrogates, choosing new runs where they disagree, and combining their predictions. Weighted combinations of surrogates are established in design optimisation [9], [10], and spectral measures of disagreement have recently been proposed for uncertainty quantification of language models [11]. Which ingredients of such schemes matter when they are used for uncertainty *evaluation* of an expensive measurement model has not been isolated.

This contribution addresses three questions. When does a passively sampled GP fail as a stand-in for an expensive model? When it fails, which ingredients of an adaptive multi-surrogate scheme restore an accurate propagation? And what does the scheme contribute to the uncertainty budget beyond a central estimate?

II. PROPOSED CONTRIBUTION

A. Prior reversion of passive GP surrogates

For a GP with prior mean μ, the posterior mean at an input $\mathbf{x}$ is

$$\hat{f}(\mathbf{x}) = \mu + \mathbf{k}_*^\top \left(\mathbf{K} + \sigma_\varepsilon^2 \mathbf{I}\right)^{-1} (\mathbf{y} - \mu\mathbf{1}), \tag{1}$$

where $\mathbf{K}$ is the covariance matrix of the training inputs, $\mathbf{k}_*$ the covariance between $\mathbf{x}$ and the training inputs, $\mathbf{y}$ the training outputs and σ_ε^2 the noise variance. When the distance from $\mathbf{x}$ to every training point greatly exceeds the kernel length scale, $\mathbf{k}_* \to \mathbf{0}$ and $\hat{f} \to \mu$: the GP reverts to a constant. We quantify when this happens with a domain-size sweep, and ask whether, and why, an adaptive committee avoids it.

B. Committee scheme under study

A committee of $K = 9$ surrogates with different inductive biases is used: two GPs, polynomial chaos expansions

Algorithm 1 Adaptive committee surrogate for propagation

Require: input PDF $g_{\mathbf{X}}$, budget N, $N_0 = 5$, acquisition a
1: $\mathcal{D} \leftarrow$ Latin hypercube of N_0 runs of f; fit all members
2: **for** $t = 1, \ldots, N - N_0$ **do**
3: $\quad \mathbf{x}^* \leftarrow \arg\max a(\mathbf{x})$ over a random candidate pool
4: $\quad$ run $f(\mathbf{x}^*)$, add it to $\mathcal{D}$, refit the members
5: $\quad$ update weights by Eq. (2), or reject members
6: **end for**
7: draw $\mathbf{X}_m \sim g_{\mathbf{X}}$, $m = 1, \ldots, M$; evaluate all members
8: **return** μ_w, σ^2_{mf}, p-box, exceedance bounds

(PCE) of degrees 2 and 3, two random forests, a support vector regressor and two neural networks. The committee is trained on a five-point Latin hypercube design, and fifteen further runs are then chosen sequentially where the members disagree most (Algorithm 1). Disagreement is scored either by a spectral measure (UEigV, the eigenvalue sum of the normalised graph Laplacian built from the K predictions [11]) or by the plain committee variance.

Two *model managers* combine the members. Hard rejection removes members that are inconsistent with the committee and averages the survivors uniformly. Error weighting assigns

$$w_k \propto \left[\mathrm{MSE}_{\mathrm{CV},k}\,(1+\pi_k)\right]^{-1}, \tag{2}$$

where $\mathrm{MSE}_{\mathrm{CV},k}$ is the cross-validation mean squared error of member k and $\pi_k \geq 0$ optionally penalises violations of known physical bounds, such as a non-negative concentration. Eq. (2) is the weighted-average surrogate of [9], [10]. We do not propose it as new; we test when it is *necessary*.

C. Propagation with a model-form component

The trained committee replaces f in a GUM S1 Monte Carlo evaluation, here with $M = 3 \cdot 10^4$ trials, which takes seconds of computing time. By the law of total variance,

$$\mathrm{Var}[Y] = \mathrm{Var}_{\mathbf{X}}\left[\mu_w(\mathbf{X})\right] + \mathrm{E}_{\mathbf{X}}\left[\sigma^2_{\mathrm{mf}}(\mathbf{X})\right], \tag{3}$$

the output variance splits into an input-driven part, carried by the weighted committee mean μ_w, and a model-form part σ^2_{mf}, the weighted spread between members. The second term's share of the total is a stopping diagnostic: further simulator runs are worthwhile while it is large. Independently of the weights, the members' output cumulative distribution functions (CDFs) span a probability box (p-box) [12], from which an exceedance probability $P(Y > y_c)$ is reported as an interval.

III. Preliminary Results and Potential Applications

A. Protocol

The test model is $f(\mathbf{x}) = \sin 2x_1 \cos 2x_2 + \mathrm{e}^{-x_3^2} + 0.2\|\mathbf{x}\|^2$, observed with noise of standard deviation 0.1, on $[-h, h]^3$ with $h = 200$ unless stated. The budget is $N = 20$ runs. Each configuration (arm) is run with 30 seeds shared by all arms. The root mean square error (RMSE) is evaluated on one fixed set of 5000 test points, and arms are compared by a two-sided Wilcoxon signed-rank test on log RMSE. A trial counts as a success if RMSE < 200; at $h = 200$ the output standard deviation (SD) is about 4100 and a prior-reverted GP reaches about 2400, so successes and failures differ by more than two orders of magnitude (Fig. 2). Code, per-seed records and the scripts that produce every number and figure are openly available [13].

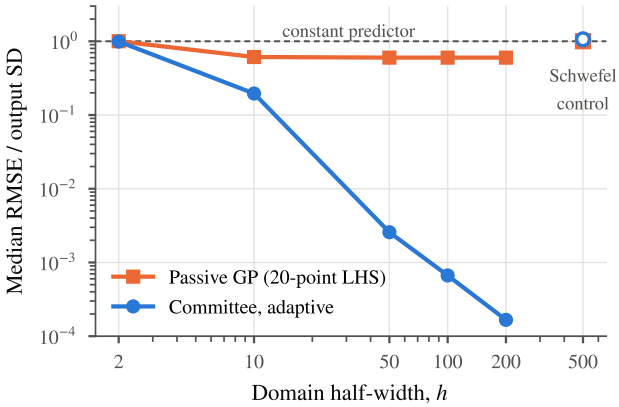


Fig. 1. Median test RMSE divided by the output SD against domain half-width (30 seeds per point). Open markers: Schwefel control on $[-500, 500]^3$. The dashed line marks a constant predictor.

B. When the passive GP fails

Fig. 1 compares a GP fitted to a 20-point Latin hypercube with the adaptive committee (UEigV, hard rejection) as the half-width h grows. The passive GP's median RMSE is 1.00 output SD at $h = 2$ and 0.60 SD for every $h \geq 10$: it reproduces little more than the mean. The committee's error falls from 0.99 SD at $h = 2$ to $1.7 \cdot 10^{-4}$ SD at $h = 200$, and the ratio of mean RMSEs from 1.04 to 0.030. The mechanism is visible in absolute terms. The quadratic term carries 44 % of the output variance at $h = 2$ and practically all of it for $h \geq 10$, and the degree-2 PCE represents it exactly. The committee's median RMSE is 0.62 to 0.68 at $h = 2$ and $h \geq 50$, close to the RMS of the oscillatory and exponential terms (0.50 to 0.61), which no member can learn from 20 runs. At $h = 10$ the median is 1.99, because hard rejection retains the degree-2 PCE in only 15 of 30 trials. On the Schwefel function on $[-500, 500]^3$, which no member represents, both methods stay at about one SD (ratio 1.10). A passive GP therefore fails whenever the run spacing is large compared with the length scale of the response, and a committee helps only to the extent that one member represents the dominant component.

C. What restores accuracy

Table I and Fig. 2 give the paired ablations at $h = 200$.

Three findings follow. *First*, the spectral metric gives no benefit: plain variance succeeds in all 30 trials and has the lower paired RMSE in 18 of 30 ($p = 0.047$). *Second*, success depends on an *anchor* member. Without the PCE members the scheme fails in every trial, as does the conventional adaptive design with a single GP. With them, hard rejection ends with the degree-2 PCE alone in 27 of 30 trials. *Third*, hard rejection collapses the committee to one member in

Table I. Paired ablations, $h = 200$, $N = 20$, 30 seeds. p: Wilcoxon signed-rank test on log *RMSE against the first row.*

Arm	Med. RMSE	Succ.	p
UEigV, hard rejection	0.682	27/30	–
Variance, hard rejection	0.660	30/30	0.047
UEigV, no PCE	2387	0/30	$< 10^{-4}$
GP alone, posterior SD	1772	0/30	$< 10^{-4}$
UEigV, Eq. (2)	0.646	29/30	0.004

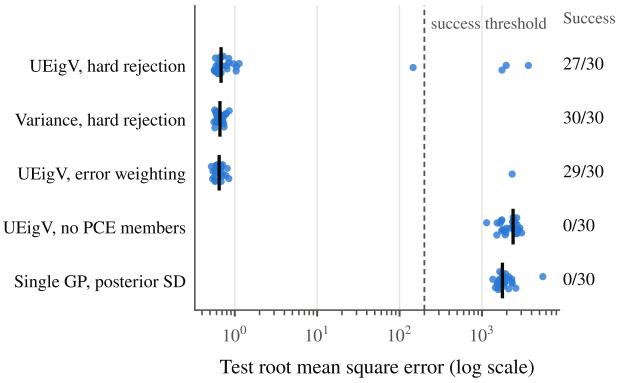


Fig. 2. Per-seed test RMSE for each arm of Table I (one dot per seed, bar at the median). Successes and failures are separated by more than two orders of magnitude.

Table II. Aggregation, $h = 200$, 30 seeds. Rows 2 and 3 score the same final survivors and data of hard rejection forced to keep $K_{\min} = 3$ members.

Manager	Aggregation	Med. RMSE	Succ.
Hard, $K_{\min} = 1$	1 left	0.682	27/30
Hard, $K_{\min} = 3$	uniform	1556	0/30
Hard, $K_{\min} = 3$	Eq. (2)	0.701	27/30
Weighting only	Eq. (2)	0.646	29/30

all 30 trials, at a median of iteration 10 of 15. After that the disagreement is identically zero, and 31 % of the run budget is silently spent on random points.

Table II isolates the aggregation rule. With $K_{\min} = 3$ the degree-2 PCE survives in 27 of 30 trials, yet the uniform mean fails in all 30, because the error of the prior-reverted members, of order 10^3, dominates it. Weighting the same survivors on the same data by Eq. (2) succeeds in all 27 trials that retain the degree-2 PCE and fails in exactly the three that do not (paired $p < 10^{-8}$). The aggregation rule, not the reversibility of rejection, is the causal ingredient, and it can only exploit an anchor that is present.

D. Propagation

Table III compares the propagated output distribution with a reference Monte Carlo evaluation of the true model ($M = 2 \cdot 10^5$) over all successful seeds. The central estimates of the two managers are indistinguishable; the difference lies in whether a model-form component can be evaluated at all.

For a representative run (seed 1000, the first seed), the weighted committee matches the reference mean to 0.21 %,

Table III. Propagation against the reference Monte Carlo evaluation, median over successful seeds (weighting: 29, hard: 27). Coverage is the share of seeds whose member envelope contains the reference value.

Quantity	Weighting	Hard
Rel. error, 5th percentile	0.68 %	0.67 %
Rel. error, median	0.23 %	0.24 %
Rel. error, 95th percentile	0.41 %	0.41 %
$\lvert\hat{P}(Y > y_{0.95}) - 0.05\rvert$	0.0015	0.0015
Width, exceedance interval	0.052	0
Coverage, mean	97 %	0 %
Coverage, exceedance	62 %	0 %
Mean p-box share of CDF	69 %	0 %

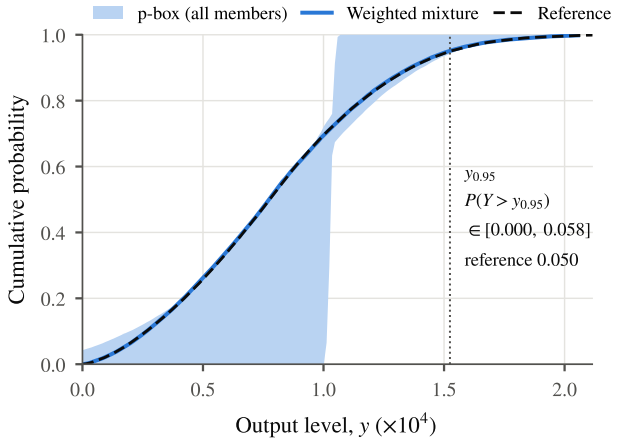


Fig. 3. Output CDF from the weighted committee (seed 1000) against the reference Monte Carlo CDF. Shaded: envelope of all nine members' CDFs; the near-vertical edge is a prior-reverted member predicting an almost constant output. Dotted: 95th percentile $y_{0.95}$.

the SD to 0.14 % and the 5th, 50th and 95th percentiles to 0.66 %, 0.24 % and 0.41 % (Fig. 3). Its model-form share in Eq. (3) falls from 0.50 at five runs to $7 \cdot 10^{-8}$ at twenty, as the weights concentrate on the degree-2 PCE. This is consistent with the actual surrogate error, whose share of the output variance is $\mathrm{RMSE}^2/\mathrm{Var}[Y] \approx 2 \cdot 10^{-8}$. The p-box and exceedance interval are unweighted member envelopes. They remain wide and are not calibrated coverage intervals: the exceedance interval contains the reference value in only 62 % of seeds. A committee reduced to one member by hard rejection gives zero-width intervals and a model-form share of exactly zero by construction, so it cannot report this component at all.

E. Potential applications

The results suggest four practical rules for using a surrogate inside a GUM S1 evaluation of an expensive model, such as a dispersion simulation in a risk assessment.

Check for prior reversion before trusting a GP. If the input ranges are wide compared with the spacing of the available runs, a passively sampled GP reproduces little more than its mean, and its own predictive variance does

not reveal this. Comparing its cross-validation error with the output SD is a cheap check.

Include members that can represent the expected structure. A committee helps only through a member whose hypothesis class matches the dominant response component, for example a low-order polynomial for a smooth trend. The choice of run-selection metric is secondary.

Weight members by out-of-sample error, never uniformly. One prior-reverted member is enough to move a uniform mean by orders of magnitude, even when an exact member is present. Cross-validation error is cheap at $N \approx 20$ and separates good and bad members by orders of magnitude.

Report the model-form component, and treat envelopes as indicative. The weighted share in Eq. (3) can be entered in the uncertainty budget alongside the input contributions; a single surrogate sets it to zero by construction, which is an omission rather than evidence of accuracy. Member envelopes show *that* the surrogate is uncertain, not with what probability; calibrated coverage needs a further step, such as conformal prediction [14].

IV. CONCLUSIONS

For an expensive measurement model evaluated about twenty times, a passively sampled GP surrogate reverts towards a constant once the run spacing is large compared with the length scale of the response. A heterogeneous committee restores accuracy only through a member that represents the dominant response component, and only if the members are combined by out-of-sample error weighting: uniform averaging failed in every trial even with that member present, and the choice of run-selection metric made no difference. The weighted committee reproduces a reference Monte Carlo evaluation to better than 1 % and makes the surrogate's own contribution visible in the uncertainty budget. These results come from a synthetic model in which an anchor member is available. The final paper will add test cases with no such member and a dispersion-type model, and will replace the uncalibrated envelopes with conformal intervals.

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